

CARLOS HOMERO JUCA VITERI

Fullstack Engineer

Loja, Ecuador · chjuca99@gmail.com · +593 99 749 8206 · linkedin.com/in/chjuca · github.com/chjuca

PROFESSIONAL SUMMARY

Software Engineer with 7 years of experience building production software with Node.js, NestJS and Python (FastAPI), and frontend with Next.js, React and Angular. Specialized in microservices, high-performance REST APIs and data access optimization using Clean Architecture and TDD. I integrate LLMs into product workflows through agents built with LangChain, LangGraph and n8n, and take end-to-end ownership of the underlying infrastructure: Kubernetes (Amazon EKS), Terraform, CI/CD and multi-account AWS operations.

EXPERIENCE

MapVX (Lazarillo) — Fullstack Developer

Dec 2021 – Present

Remote

- Designed REST APIs and microservices in Node.js / NestJS and Python / FastAPI for route precomputation and geospatial search, **cutting response time from 5,000 ms to 900 ms** while integrating third-party geospatial services without degrading performance.
- Built agents and automations with LangChain, LangGraph and n8n integrated with internal APIs, automating operational processes that were previously manual.
- Developed features in Next.js, React and Angular 18 for the web app and internal indoor-map drawing tools (Mapbox GL, Leaflet), applying Clean Architecture and TDD to reduce recurring production bugs.
- Optimized critical queries in PostgreSQL, MongoDB and DocumentDB through indexing and query refactoring, removing backend bottlenecks.
- Operated production infrastructure across 3 AWS accounts (EKS, ECS, S3, CloudFront, Lambda, RDS) with Kubernetes, Helm and Terraform, and built CI/CD pipelines with GitHub Actions and Docker that eliminated manual deployments.
- Implemented monitoring with Elastic Stack, CloudWatch and Uptime Kuma, and managed Nginx as a reverse proxy for ~87 virtual hosts along with the TLS certificate lifecycle, resolving incidents with zero downtime for clients.

ExmerDev — Backend Developer (Java · Spring Boot)

Sep 2020 – Nov 2021

Loja, Ecuador

- Implemented the roles, users and security modules of a Trading and CRM system with Spring Boot and Spring Cloud, integrating asynchronous messaging with RabbitMQ to decouple services.
- Optimized MySQL, PostgreSQL and MongoDB databases and documented the team's REST endpoints, reducing integration friction.

Teaching Innovation and Research Lab — Backend Lead

Apr 2019 – Aug 2020

Universidad Técnica Particular de Loja · Loja, Ecuador

- Developed a REST API with Node.js and PostgreSQL to manage attendance, people and materials, digitizing a manual process.
- Coordinated backend development of the LiiD Manager platform (health and education) and defined technical standards and code reviews for the team.

TECHNICAL SKILLS

- **Backend:** Node.js, NestJS, Express, TypeScript, Python, FastAPI, Java, Spring Boot, Microservices, REST, RabbitMQ
- **Frontend:** Next.js, React, Angular, Mapbox GL, Leaflet
- **AI & Automation:** LangChain, LangGraph, n8n, LLM integration
- **Databases:** PostgreSQL, MySQL, MongoDB, DocumentDB, Firebase
- **Cloud & DevOps:** AWS, Kubernetes, Helm, Terraform, Docker, GitHub Actions, Azure, GCP, Nginx, Linux
- **Observability & practices:** Elastic Stack, CloudWatch, Uptime Kuma, Clean Architecture, TDD, Jest, Git, Scrum

EDUCATION

B.S. in Computer Systems Engineering

2017 – 2022

Universidad Técnica Particular de Loja · Loja, Ecuador

LANGUAGES

Spanish · English